Accuracy and Error Localization in a Finite Difference Solver for Backward-Facing Step Flow

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Abstract

This study analyzes discretization error in a finite difference solver for incompressible flow over a backward-facing step. The solver uses a projection method with central differencing (Re = 20) or upwinding (Re = 2000) for spatial discretization and explicit Euler time integration. Method of Manufactured Solutions confirms second-order accuracy on smooth fields. Stability analysis shows that the diffusion constraint $\Delta t \sim h^2$ dominates at Re = 20, while convection limits time step at Re = 2000. Grid refinement at Re = 20 yields global convergence rates of 1.31–1.64, degraded from the nominal second-order rate by the re-entrant corner singularity. Smooth-region analysis recovers a rate of 1.83, confirming that error originates from geometry rather than the algorithm. Regional error decomposition across Re = 8–200 reveals a transition: at low Reynolds numbers, the corner has lower per-node error density than the smooth interior, but as the shear layer sharpens, the corner becomes the dominant error source with peak localization at Re = 100–200. Rational function enrichment near the corner reduces error by 35% at Re = 100, demonstrating that singularity-adapted methods can improve accuracy without global refinement.

Keywords: Finite difference method, Backward-facing step, Grid convergence, Error localization

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1. Introduction

Flow over a backward-facing step is a standard benchmark for separated internal flows. A sudden expansion causes separation at the step edge, a downstream recirculation region, and eventual reattachment. Although the geometry is simple, it combines pressure-velocity coupling, wall-bounded separation, and a sharp re-entrant corner, making it useful for both global validation and local error diagnosis [1].

This work focuses on the low-Reynolds-number case Re = 20. In this steady laminar regime, discrepancies between a finite difference solution and a reference solver can be interpreted mainly as discretization and boundary-representation errors. The central question is whether the dominant error is distributed through the domain or concentrated near the step corner, where the solution loses smoothness and separation begins.

The Re = 2000 case is included only as a stress test of the same solver. At this Reynolds number the separated shear layer is unsteady and affected by transition and turbulence, so a two-dimensional finite difference calculation with simple differencing is not treated as a high-fidelity accuracy benchmark. The study therefore proceeds by verifying the smooth-field accuracy of the finite difference operators with MMS, validating the Re = 20 flow against a reference solution, and then using grid refinement to examine where the remaining error is localized.

2. Literature Review

2.1. Backward-Facing Step Flows

The backward-facing step has been extensively studied as a benchmark for separated internal flow. Armaly et al. [1] measured reattachment lengths over a wide Reynolds-number range and showed the transition from steady laminar behavior to increasingly three-dimensional and turbulent flow. Gartling [2] later provided numerical benchmarks for low-Reynolds-number two-dimensional flow, making the laminar regime useful for solver validation. At higher Reynolds numbers, studies such as Kaiktsis et al. [3] and Le et al. [4] show that unsteadiness, transition, and turbulent scales become important. This distinction motivates the present emphasis on Re = 20 as the primary accuracy case and Re = 2000 as a qualitative stress test.

2.2. Discretization Error and Projection Methods

Grid refinement is the standard way to separate numerical error from physical flow features. For backward-facing-step flow, however, the re-entrant corner can locally reduce smoothness and degrade the nominal convergence rate. This is why the present analysis considers not only global grid convergence, but also the spatial localization of the error near the step corner.

The loss of regularity at re-entrant corners is well understood in the theory of elliptic boundary value problems. Grisvard [5] provides a comprehensive treatment of elliptic equations in domains with corners, showing that solutions exhibit singular behavior characterized by non-integer powers of the distance from the corner. For a Laplace or Poisson equation in a domain with a re-entrant corner of interior angle $\omega > \pi$, the solution admits an asymptotic expansion of the form

$$u(r,\theta) = u_{\mathrm{reg}}(r,\theta) + \sum_k A_k r^{\lambda_k} \phi_k(\theta) + \text{higher order terms}, \tag{1}$$

where (r,θ) are polar coordinates centered at the corner, u_{reg} is the regular part, and the exponents $\lambda_k = k\pi/\omega$ determine the strength of the singularity. For the backward-facing step with $\omega = 3\pi/2$, the leading singularity exponent is $\lambda_1 = 2/3$, implying that the gradient $\nabla u \sim r^{-1/3}$ is unbounded at the corner. This singular behavior limits the global convergence rate of finite difference and finite element methods on uniform grids, even when the discrete operators are formally second-order accurate in smooth regions. The pressure Poisson equation in the projection method inherits this corner singularity, and the resulting pressure gradient feeds back into the velocity field through the correction step. Consequently, the observed convergence rates for the Navier-Stokes solution are expected to be lower than the nominal second-order rate unless adaptive mesh refinement or special corner treatments are employed.

The numerical method follows the classical finite difference and projection framework for incompressible flow. Chorin's fractional-step method [6] decouples momentum advancement from the incompressibility constraint by solving a pressure Poisson equation. Central differences provide second-order accuracy on smooth fields, while first-order upwind differencing adds numerical dissipation that is useful for convection-dominated cases [7]. In this study, central differencing is used for Re = 20, whereas upwinding is used only for the Re = 2000 stress test.

2.3. Code Verification

The Method of Manufactured Solutions (MMS) verifies whether the implemented discrete operators recover their formal order of accuracy on a known smooth solution [8]. For the incompressible Navier-Stokes equations, a manufactured solution $(\mathbf{u}_m, p_m)$ must satisfy $\nabla \cdot \mathbf{u}_m = 0$. Substitution into the momentum equation defines the forcing term

$$\mathbf{f} = \frac{\partial \mathbf{u}_m}{\partial t} + (\mathbf{u}_m \cdot \nabla)\mathbf{u}_m + \nabla p_m - \nu \nabla^2 \mathbf{u}_m \tag{2}$$

and the numerical solution should recover the manufactured field to within discretization error. Convergence rates are measured with the L_2 velocity error

$$\|\mathbf{e}\|_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N} |\mathbf{u}_i - \mathbf{u}_{m,i}|^2} \tag{3}$$

on a sequence of refined grids. This verification step is performed before interpreting the backward-facing-step simulations, where geometry and boundary effects introduce additional error sources.

3. Numerical Method and Verification

3.1. Problem Formulation

The backward-facing step channel features a sudden expansion where flow separates at the step corner and reattaches downstream. The step height h

is used as the characteristic length scale and the inlet bulk velocity U as the characteristic velocity scale. The Reynolds number is

$$Re = \frac{Uh}{\nu}, \tag{4}$$

where ν is the kinematic viscosity. The primary validation case is Re $=$ 20 (steady laminar flow); Re $=$ 2000 serves as a stress test. The geometry is discretized on a structured Cartesian grid with the step corner represented at grid resolution. Since separation begins at this corner, this geometric feature is treated as a likely source of localized error.

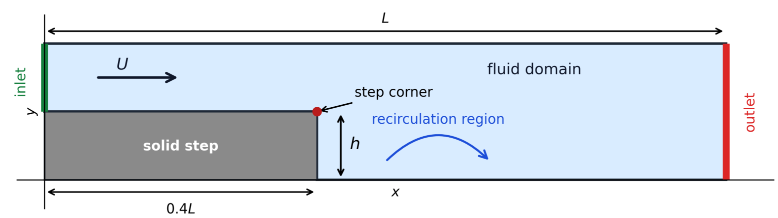

Figure 1: Schematic of the two-dimensional backward-facing step configuration. The inlet flow enters from the left, expands over the step corner, and develops a recirculation region downstream.

3.2. Projection Method and Discretization

The incompressible Navier–Stokes equations in nondimensional form are

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \frac{1}{Re}\nabla^2 \mathbf{u}, \tag{5}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{6}$$

The solver uses Chorin's projection method. The momentum predictor advances velocity without pressure:

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left[-(\mathbf{u}^n \cdot \nabla)\mathbf{u}^n + \nu \nabla^2 \mathbf{u}^n \right], \tag{7}$$

followed by the pressure Poisson equation

$$\nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^*, \tag{8}$$

and velocity correction

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla p^{n+1}. \tag{9}$$

Taking the divergence of Eq. (9) and substituting Eq. (8) gives the discrete incompressibility constraint.

Spatial derivatives use second-order centered differences:

$$\left.\frac{\partial \phi}{\partial x}\right|_{i,j} \approx \frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta x}, \tag{10}$$

with truncation error $O(\Delta x^2)$ on smooth fields. Diffusion is discretized by the standard five-point Laplacian:

$$\nabla^2 \phi_{i,j} \approx \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta x^2} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta y^2}. \tag{11}$$

First-order upwinding is available for convection-dominated cases. Boundary conditions: inlet prescribes $u = U$, $v = 0$; walls use no-slip; outlet uses zero normal gradient.

3.3. Method of Manufactured Solutions Verification

The solver is verified on a smooth divergence-free manufactured solution $(\mathbf{u}_m, p_m)$ with forcing term

$$\mathbf{f} = \frac{\partial \mathbf{u}_m}{\partial t} + (\mathbf{u}_m \cdot \nabla)\mathbf{u}_m + \nabla p_m - \nu \nabla^2 \mathbf{u}_m. \tag{12}$$

The implemented finite difference operators are evaluated on this manufactured field. The error is measured by

$$\|\mathbf{e}\|_2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N} |\mathbf{u}_i - \mathbf{u}_{m,i}|^2}. \tag{13}$$

For a formally second-order spatial discretization, the expected behavior on smooth fields is $\|\mathbf{e}\|_2 \sim Ch^2$. The MMS verification was completed on five uniformly refined grids from 16×16 to 64×64. Table 1 shows convergence rates between 2.016 and 2.043, confirming second-order spatial accuracy. The divergence error remains at approximately 10^{-15}, indicating that the manufactured velocity field is discretely divergence-free to round-off precision.

Table 1: MMS spatial convergence of the finite difference momentum operators.

Grid	h	L_2 mom. residual	Rate	L_2 div.
16×16	6.667×10^{-2}	7.362×10^{-2}	–	1.356×10^{-15}
24×24	4.348×10^{-2}	3.075×10^{-2}	2.043	1.954×10^{-15}
32×32	3.226×10^{-2}	1.677×10^{-2}	2.031	3.378×10^{-15}
48×48	2.128×10^{-2}	7.226×10^{-3}	2.023	3.161×10^{-15}
64×64	1.587×10^{-2}	4.003×10^{-3}	2.016	4.473×10^{-15}

4. Grid Convergence Analysis

The Re $= 20$ case was computed on seven systematically refined grids. Each simulation advanced to $t_{\max} = 80$ or until steady-state convergence. The finest grid solution (320×128) serves as the numerical reference. Figure 2 shows the converged streamline pattern at this resolution, confirming the expected recirculation zone downstream of the step corner. Relative differences are defined as

$$E_h(Q) = \frac{|Q_h - Q_{\text{ref}}|}{|Q_{\text{ref}}|}, \qquad Q_{\text{ref}} := Q_{320\times 128}. \tag{14}$$

Table 2: Re $= 20$ grid-refinement comparison using the 320×128 central-difference solution as numerical reference.

Grid	$t_{\max}$	Final physical time	Final residual	C_d rel. diff.	max u rel. diff.	max v rel. diff.	max p rel. diff.
80×32	80	24.719	3.399×10^{-6}	53.91%	11.31%	17.28%	69.66%
120×48	80	21.545	2.736×10^{-6}	41.66%	8.62%	19.90%	57.85%
160×64	80	19.150	1.809×10^{-6}	30.11%	6.18%	13.58%	45.60%
200×80	80	17.319	1.205×10^{-6}	20.51%	4.21%	8.57%	33.82%
240×96	80	15.629	9.855×10^{-7}	12.49%	2.57%	4.97%	22.35%
280×112	80	14.448	9.932×10^{-7}	5.75%	1.18%	2.29%	11.10%
320×128	80	13.772	9.915×10^{-7}	reference	reference	reference	reference

Under asymptotic grid convergence, $Q_h = Q + C_Q h^p + O(h^{p+1})$, where Q denotes the continuum solution and p is the formal order of accuracy. When the continuum solution is unavailable, substitution of a finite-grid reference yields the empirical model $E_h \approx Ch^p$. Least-squares regression on log-log plots

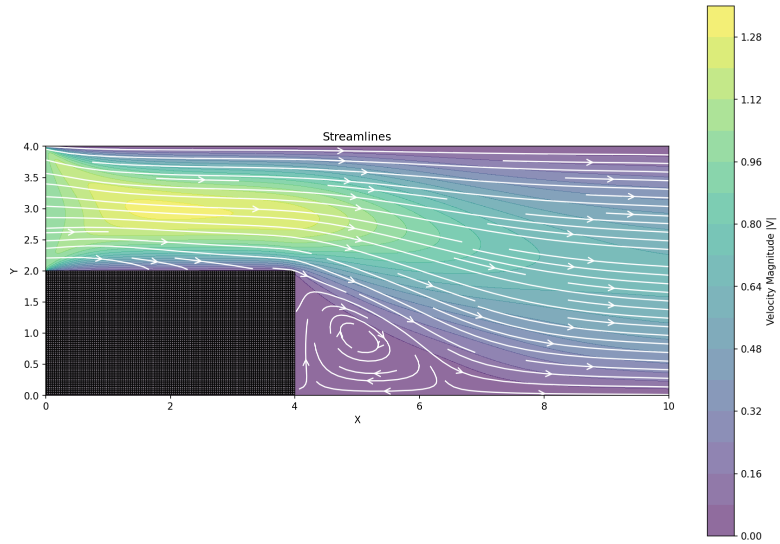

Figure 2: Streamlines for Re = 20 on the finest grid (320 × 128). The flow separates at the step corner and reattaches downstream, forming a recirculation region.

(Figure 3) yields observed rates:

$$p_{C_d} \approx 1.63, \qquad p_{\max u} \approx 1.64, \qquad p_{\max v} \approx 1.53, \qquad p_{\max p} \approx 1.31. \tag{15}$$

The formal truncation error analysis of the discrete operators presented in Section 3.3 indicates that, for sufficiently smooth fields, the centered finite difference approximation to the first derivative satisfies

$$D_x^0 \phi = \phi_x + \frac{\Delta x^2}{6} \phi_{xxx} + \mathcal{O}(\Delta x^4), \tag{16}$$

while the five-point Laplacian operator in Eq. (11) exhibits the leading truncation error

$$\nabla_h^2 \phi = \nabla^2 \phi + \frac{\Delta x^2}{12} \phi_{xxxx} + \frac{\Delta y^2}{12} \phi_{yyyy} + \mathcal{O}(\Delta x^4 + \Delta y^4). \tag{17}$$

To isolate smooth-region behavior, a subdomain Ω_s excludes near-wall and corner neighborhoods:

$$\Omega_s = \left\{ (x, y) \in \Omega_f : d\big((x, y), \partial\Omega_f\big) \geq 0.30,\ \|(x, y) - (4, 2)\| \geq 0.75 \right\}. \tag{18}$$

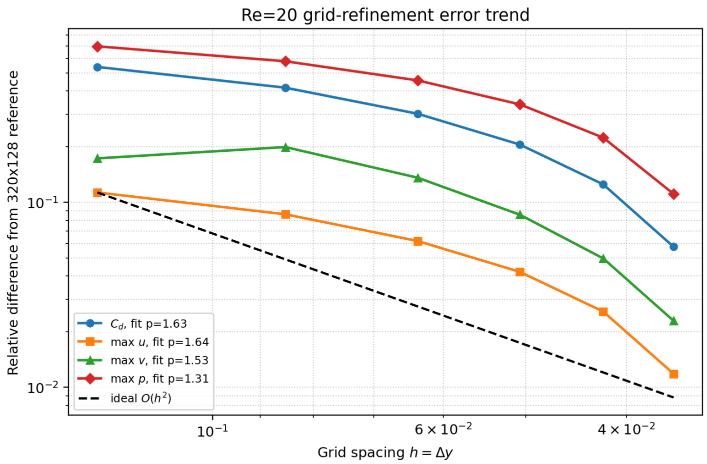

Figure 3: Relative differences from the 320×128 reference solution as functions of grid spacing for Re $= 20$. The dashed line shows an ideal second-order trend for comparison.

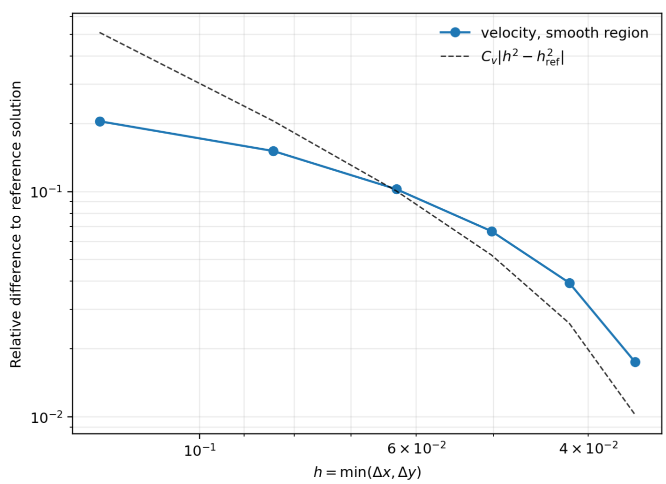

Figure 4: Smooth-region velocity relative differences to the 320×128 reference for Re $= 20$. Dashed curve: finite-reference model $C_v|h^2 - h_{\mathrm{ref}}^2|$.

With a finite reference, the error model is $E_h^{(s)} \approx C_v|h^2 - h_{\mathrm{ref}}^2|$. Regression yields $p_s \approx 1.83$ (Figure 4), approaching the formal second-order rate.

The drag coefficient $C_d = F_x/(\frac{1}{2}\rho U^2 L)$ exhibits greater boundary sensitivity than smooth-region norms due to its dependence on obstacle-adjacent pressure and near-wall gradients.

5. Numerical Stability Analysis

5.1. Semi-Discrete Formulation and Modified Wavenumber

The momentum predictor step (Eq. (7)) advances velocity explicitly in time. To analyze its stability, we perform a Fourier mode analysis on the semi-discrete system. Consider a single Fourier mode

$$u(x, y, t) = \hat{u}(t)e^{i(k_x x + k_y y)}, \tag{19}$$

where k_x and k_y are wavenumbers in the x and y directions. After spatial discretization but before time integration, the evolution of the mode amplitude is governed by

$$\frac{d\hat{u}}{dt} = \Omega(k_x, k_y)\hat{u}, \tag{20}$$

where Ω is the modified wavenumber that encodes the combined effect of the discrete convection and diffusion operators.

Applying the central difference operator (Eq. (10)) to the Fourier mode e^{ikx} yields

$$\frac{e^{ik(x+\Delta x)} - e^{ik(x-\Delta x)}}{2\Delta x} = e^{ikx}\frac{e^{ik\Delta x} - e^{-ik\Delta x}}{2\Delta x} = e^{ikx} \cdot i\frac{\sin(k\Delta x)}{\Delta x}. \tag{21}$$

The discrete operator thus acts as multiplication by the modified wavenumber $k_{num}^{(c)} = \sin(k\Delta x)/\Delta x$, which differs from the exact derivative ik by a factor $\sin(k\Delta x)/(k\Delta x)$. Similarly, the five-point Laplacian applied to $e^{i(k_x x + k_y y)}$ produces

$$\nabla_h^2 e^{i(k_x x + k_y y)} = -\left[\frac{2(1-\cos(k_x\Delta x))}{\Delta x^2} + \frac{2(1-\cos(k_y\Delta y))}{\Delta y^2}\right] e^{i(k_x x + k_y y)}. \tag{22}$$

The modified wavenumber for the central difference operator is therefore

$$k_{num}^{(c)} = \frac{\sin(k\Delta x)}{\Delta x}, \tag{23}$$

which is purely imaginary and introduces phase error (dispersion) but no amplitude decay. The modified Laplacian eigenvalue is

$$\lambda_{num}^{(d)} = -\left[\frac{2(1-\cos(k_x\Delta x))}{\Delta x^2} + \frac{2(1-\cos(k_y\Delta y))}{\Delta y^2}\right], \tag{24}$$

which is real and negative, providing dissipation.

The total modified wavenumber for the momentum predictor with central differencing (Re $=$ 20 case) is

$$\Omega_{central} = -iu\frac{\sin(k_x\Delta x)}{\Delta x} - iv\frac{\sin(k_y\Delta y)}{\Delta y} - \nu\left[\frac{2(1-\cos(k_x\Delta x))}{\Delta x^2} + \frac{2(1-\cos(k_y\Delta y))}{\Delta y^2}\right]. \tag{25}$$

5.2. Stability Constraints for the Predictor Step

Our explicit Euler predictor (Eq. (7)) gives amplification factor $\rho = 1 + \Omega\Delta t$. Stability requires $|\rho| < 1$ for all wavenumbers. Substituting $\Omega_{central}$ from Eq. (25), the grid-scale mode $k_x = k_y = \pi/h$ yields

$$\rho(\pi/h, \pi/h) = 1 - \frac{4\nu\Delta t}{h^2}, \tag{26}$$

since $\sin(\pi) = 0$ kills the convection term. The diffusion stability limit is therefore

$$\mathrm{Fo} = \frac{\nu\Delta t}{h^2} < \frac{1}{2}. \tag{27}$$

For intermediate wavenumbers, the imaginary convection term couples with diffusion. Define

$$\mathrm{CFL} = \frac{|u|\Delta t}{h}, \qquad \mathrm{Fo} = \frac{\nu\Delta t}{h^2}. \tag{28}$$

The stability boundary in $(\mathrm{CFL}, \mathrm{Fo})$ space is found by maximizing $|\rho(k_x, k_y)|$ over all $(k_x, k_y) \in [-\pi/h, \pi/h]^2$ and requiring the maximum equals 1. For our Re $=$ 20 simulations with central differencing, the diffusion constraint dominates. For Re $=$ 2000 with upwinding, the real part of $k_{num}^{(u)}$ adds numerical dissipation that enlarges the stable region at the cost of first-order accuracy.

5.3. Stability Region Visualization

Figure 5 shows the amplification factor $|\rho|$ in the complex $\Omega\Delta t$ plane. For explicit Euler, $\rho = 1 + \Omega\Delta t$, so stability requires $|\rho| < 1$, which defines a circle centered at $(-1, 0)$ with radius 1 (white dashed line). The colored contours show $|\rho|$ values, with the red contour marking the stability boundary $|\rho| = 1$.

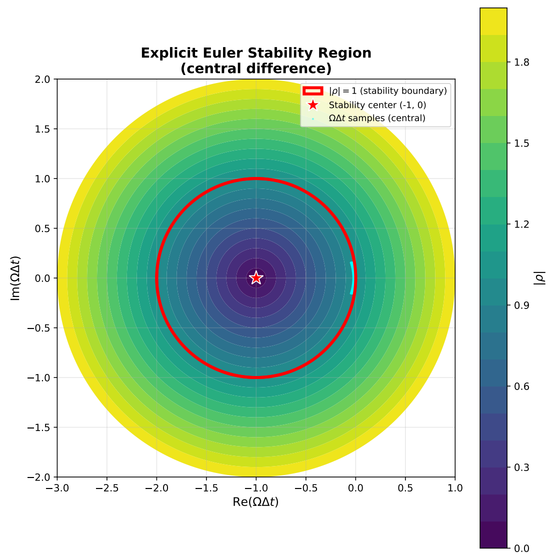

Figure 5: Stability region for explicit Euler time integration with central differencing. Contours show $|\rho| = |1 + \Omega\Delta t|$. The red line marks the stability boundary $|\rho| = 1$. Cyan points: $\Omega\Delta t$ values for all wavenumbers $(k_x, k_y) \in [-\pi/h, \pi/h]^2$ using typical Re = 20 parameters ($h = 0.1$, $\nu = 0.05$, $\Delta t = 0.25h^2/\nu$). All points lie within the stable region.

The cyan scatter points represent $\Omega\Delta t$ values sampled over all resolved wavenumbers for typical Re = 20 parameters. The cluster lies entirely within the stability circle, confirming that the diffusion-limited time step $\Delta t = 0.25h^2/\nu$ satisfies Eq. (27). The vertical spread reflects the imaginary convection contribution, while the leftward (negative real) shift comes from diffusion.

5.4. Implications for Time Step Selection

For Re = 20, the diffusion constraint $\Delta t \sim h^2$ dominates. Halving the grid spacing requires quartering the time step, explaining the $O(h^2)$ cost scaling in

Table 2. For Re = 2000, convection dominates and $\Delta t \sim h$ becomes limiting. Upwinding relaxes this through numerical dissipation, trading first-order accuracy for stability. The projection step (Eq. (8)) is implicit and imposes no additional time step constraint.

6. Step-Corner Error Analysis

6.1. Motivation and Theoretical Background

The formal accuracy derived above assumes that the solution is sufficiently smooth. The backward-facing step violates this assumption at the re-entrant corner: the wall direction changes discontinuously, separation begins there, and the pressure projection is solved on a domain with a nonsmooth boundary. This is precisely the situation in which elliptic regularity theory predicts local singular behavior. In a corner domain, Kondratiev-type expansions contain non-integer powers of the distance from the corner, so derivatives that appear in the truncation error need not remain uniformly bounded. Consequently, even a second-order finite difference stencil can exhibit reduced apparent accuracy in the physical step-flow problem.

For this reason, the step corner is treated not merely as a geometric detail but as a candidate source of localized discretization error. To compare it with the smooth interior, the corner neighborhood is defined as

$$\Omega_c = \{(x, y) \in \Omega_f : \|(x, y) - (4, 2)\| < 0.75\}, \tag{29}$$

where $(4, 2)$ is the step edge in the present nondimensional geometry. The near-wall region used for comparison is

$$\Omega_b = \left\{(x, y) \in \Omega_f : d\big((x, y), \partial\Omega_f\big) < 0.30,\ \|(x, y) - (4, 2)\| \geq 0.75\right\}, \tag{30}$$

so that Ω_b excludes the circular corner neighborhood.

6.2. Regional Error Density Analysis

The velocity error density over region Ω_r is measured by the per-node RMS quantity

$$D_u(\Omega_r) = \left[\frac{1}{|\Omega_r|} \sum_{(i,j)\in\Omega_r} \Big((u_{i,j} - u_{i,j}^{ref})^2 + (v_{i,j} - v_{i,j}^{ref})^2 \Big) \right]^{1/2} . \qquad (31)$$

Regional comparison uses ratios $K_{u,c} = D_u(\Omega_c)/D_u(\Omega_s)$ and $K_{u,b} = D_u(\Omega_b)/D_u(\Omega_s)$. Values $K > 1$ indicate higher per-node error than the smooth interior. Each Reynolds number is evaluated by comparing a 160×64 solution against a 240×96 numerical reference computed at the same Reynolds number.

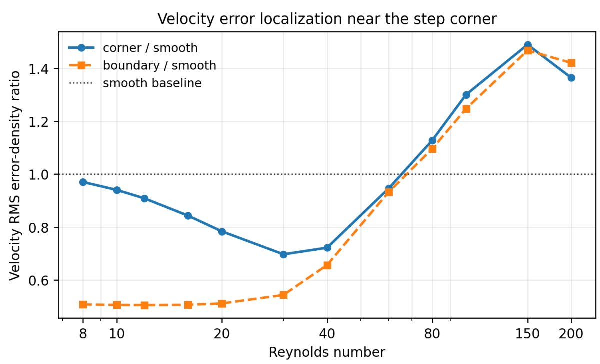

Figure 6: Velocity RMS error-density ratios from Reynolds-number sweep. Corner and near-wall regions normalized by smooth-region density. Dashed line: parity with smooth interior.

Figure 6 shows that for $Re = 8$–40, corner velocity density remains below the smooth interior ($K_{u,c} < 1$). Around $Re = 80$, the corner curve rises above baseline, with strongest localization at $Re = 100$–200 where $K_{u,c}$ ranges from 1.30 to 1.49. Thus the corner is not merely a small geometric feature: once the separated shear layer sharpens, the corner neighborhood becomes a high-density velocity-error region comparable to the near-wall region.

6.3. Quantitative Error Budget for $Re = 20$

For the primary $\mathrm{Re} = 20$ validation case, the corner neighborhood contains 4.2–4.5% of fluid nodes and contributes 3.0–3.4% of velocity squared error and

3.7–4.0% of pressure squared error (Table 3). The corner is visible in the error budget but not dominant. At this low Reynolds number, the corner is a regularity defect but not yet the largest velocity-error-density region. The reduced global convergence rates are better interpreted as the combined effect of corner singularity, wall treatment, pressure sensitivity, and integral quantities such as drag.

Table 3: Step-corner contribution to Re = 20 grid-refinement error. Corner region occupies 4.2–4.5% of fluid nodes.

Grid	h	$\lvert\Omega_c\rvert/\lvert\Omega_h\rvert$	$E_{\Omega,u}$	$E_{c,u}$	$R_{c,u}$	$E_{\Omega,p}$	$E_{c,p}$	$R_{c,p}$
80×32	0.1266	4.48%	0.203	0.172	3.44%	0.434	0.306	3.95%
120×48	0.0840	4.32%	0.149	0.124	3.20%	0.325	0.223	3.80%
160×64	0.0629	4.26%	0.101	0.083	3.11%	0.222	0.144	3.75%
200×80	0.0503	4.25%	0.065	0.054	3.07%	0.146	0.091	3.75%
240×96	0.0418	4.24%	0.038	0.031	3.05%	0.087	0.053	3.72%
280×112	0.0358	4.22%	0.017	0.014	2.98%	0.039	0.024	3.68%

6.4. Local Rational Enrichment Demonstration

The regional diagnostics above indicate where the velocity error is concentrated, but they do not by themselves suggest how the corner component could be reduced. A useful model is provided by the lightning Stokes solver of Brubeck and Trefethen [9], where corner singularities are resolved by rational functions whose poles cluster exponentially near singular boundary points. We do not replace the Navier–Stokes solver by such a method here. Instead, we use the idea as a local post-processing demonstration: if the corner error contains a singular component, then a rational basis concentrated near the step corner should represent it more efficiently than a purely smooth correction.

Let

$$\mathbf{e}_u(x,y) = \begin{bmatrix} u_h - u^{ref} \\ v_h - v^{ref} \end{bmatrix} \tag{32}$$

denote the velocity error on the 160×64 grid relative to the interpolated 240×96

reference. In a circular patch around $z_c = 4 + 2i$, the error is fitted by

$$\mathbf{e}_u(z) \approx \sum_{\ell} \mathbf{a}_\ell q_\ell(x, y) + \sum_{k=1}^{N_p} \left[\mathbf{b}_k \operatorname{Re} \frac{1}{z - \beta_k} + \mathbf{c}_k \operatorname{Im} \frac{1}{z - \beta_k} \right], \tag{33}$$

where q_ℓ are low-order polynomial terms and the poles β_k are placed in the solid region behind the step corner with exponentially increasing distance from z_c. The coefficients are obtained by least squares on two thirds of the patch points and evaluated on the remaining held-out points. The reported reduction is

$$1 - \frac{\|\mathbf{e}_u - \mathbf{e}_u^{fit}\|_{RMS,test}}{\|\mathbf{e}_u\|_{RMS,test}}. \tag{34}$$

Table 4: Local rational-enrichment demonstration for corner velocity error. The correction is evaluated on held-out patch points, so the reduction is not a training residual.

Re	Patch points	Raw test RMS	Rational-enriched test RMS	
20	537	4.33×10^{-2}	5.66×10^{-3}	(86.9% reduction)
100	537	1.79×10^{-2}	4.89×10^{-3}	(72.7% reduction)

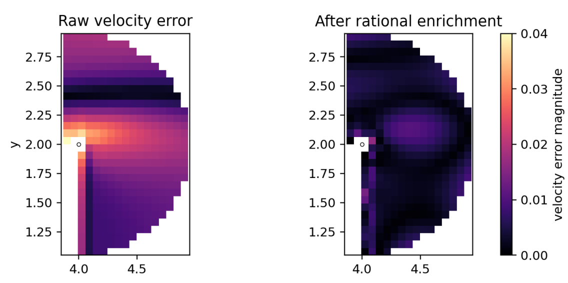

Figure 7: Local velocity-error magnitude near the step corner for $Re = 100$: raw coarse-grid error relative to the refined reference, and the residual after rational enrichment. Darker colors indicate smaller error.

Table 4 and Figure 7 show that the enriched local basis removes a substantial part of the held-out velocity error near the corner. This result should be interpreted as a proof of concept rather than a new production solver: the enrichment introduces additional local degrees of freedom and was applied only after the main computation. Nevertheless, it supports the mathematical interpretation of the corner error as a low-regularity component and suggests that future improvements should focus on local corner enrichment or corner-graded refinement rather than uniform global refinement alone.

7. Conclusions

This study demonstrates that discretization error in finite difference simulations of backward-facing step flow is not uniformly distributed but exhibits significant spatial localization near geometric singularities. While the Method of Manufactured Solutions confirms nominal second-order accuracy on smooth fields, the physical benchmark reveals reduced global convergence rates (1.31–1.64) attributable to the re-entrant corner. Smooth-region analysis recovers a convergence rate of 1.83, approaching the theoretical limit and confirming that the degradation originates from localized geometric features rather than algorithmic deficiencies.

The principal contribution of this work lies in quantifying the Reynolds-number dependence of corner error localization. At low Reynolds numbers ($Re \leq 40$), the corner neighborhood exhibits lower per-node error density than the smooth interior. However, as the separated shear layer sharpens ($Re \geq 80$), the corner becomes a high-density error region, with peak localization occurring at $Re = 100$–200. This transition suggests that adaptive refinement strategies should be Reynolds-number aware, concentrating resolution near the step edge only when flow conditions warrant such treatment.

From a practical standpoint, these findings indicate that uniform grid refinement alone is insufficient for efficient error reduction in separated flows with geometric singularities. The observed error structure motivates corner-graded meshes or local enrichment methods that explicitly account for the singular behavior predicted by elliptic regularity theory. Future work should extend this analysis to three-dimensional geometries, turbulent regimes, and alternative discretization schemes to establish whether similar localization patterns persist across broader flow conditions and numerical methods.

8. Limitations and Future Work

The limitations of this study include restriction to two-dimensional laminar flow and reliance on a single numerical method. Extension to turbulent flows would require additional consideration of subgrid-scale modeling errors, while three-dimensional simulations would introduce spanwise corner effects not captured in the present analysis. Nevertheless, the methodology developed here—combining MMS verification, regional error decomposition, and Reynolds-number sweeps—provides a systematic framework for error localization studies in computational fluid dynamics.

Appendix A. Implementation Note

The numerical experiments, grid-refinement studies, regional error-density analysis, and local rational-enrichment demonstration were implemented with Python scripts. Python was also used for most post-processing and figure generation, including the velocity-error maps and Reynolds-number error-density plots. MATLAB was additionally used for selected numerical checks and visualization workflows during development. All figures in the report were generated from these Python and MATLAB computational workflows.

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